ic_autolab round 5 (generic track): report

Protocol: PROTOCOL.md, frozen 2026-11-04T04:14:21Z at git `fafd6d1ce992609429076251a3d2a97eb847b0bd`; route `local`. Counted GAE only (deterministic, host independent); no wall-time claim; development loop, no validation-gate row. Loop home: `../`; ledger: `../ledger.json`.

Claim record: `PCC-20261104-0213f5` (`python3 tools/pcc/pcc.py standing PCC-20261104-0213f5`). Until a signed replay and recompute exist, this round's outcome is an unverified assertion [pcc:unverified].

Outcome

k	class	sel	sel / inc at w = 0 [95%]	w = 0.1	w = 1	band	S sel (w = 0 / 0.1 / 1)	S inc	S rho-neg	distance sel / rho
32	**improvement**	ksum4-n26	0.352 [0.340, 0.363]	0.341 [0.329, 0.353]	0.291 [0.276, 0.306]	sel < inc	8.296 / 8.811 / 13.446	23.597 / 25.866 / 46.285	0.911 / 1.001 / 1.806	9.1x / 8.8x / 7.4x

Classes follow the protocol's decision rule. Distance to rho is the distance remaining, not a gate. The reasons per size:

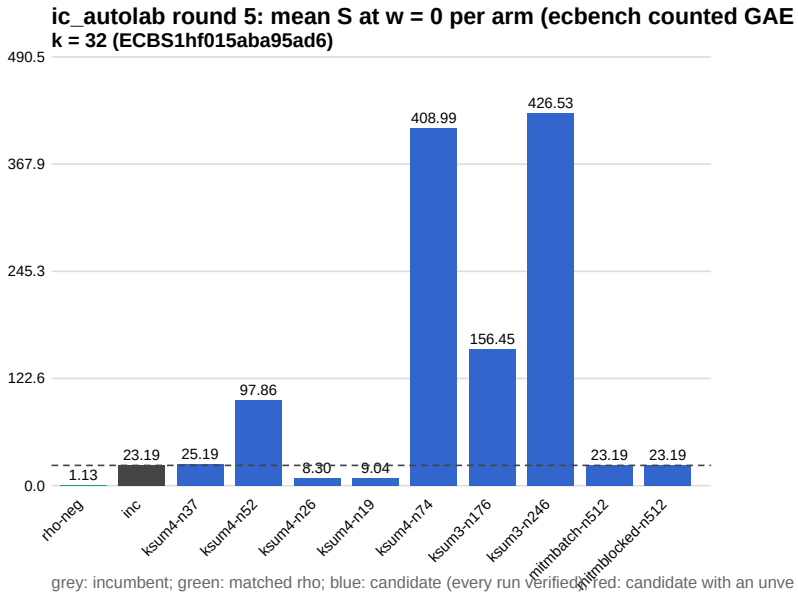
- k = 32: compare ops.status = ok, admissible = true; sel / inc = 0.3516 [0.340, 0.363] at w = 0; band: sel < inc; A/A inc-aa / inc interval [1.000, 1.000] contains 1

Train (development screen, in sample)

k = 32: session `ECBS1hf015aba95ad6` (run exit 0, verify exit 0, receipt sha256 `f5016f5397959d4d5aff48bc5a9398e850511d76219dd7a1000dc73475700790`)

arm	family	N	factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`rho-neg`	rho	-	-	`rho.negation`	16	16	1.135 / 1.247 / 2.253	matched reference
`inc`	mitm2	512	`prime-projected-abscissa:size=512,scan_cap=8192`	`mitm:negation_folded=1`	16	16	23.193 / 25.381 / 45.069	incumbent
`ksum4-n37`	ksum4	37	`prime-projected-abscissa:size=37,scan_cap=592`	`ksum:m=4`	16	16	25.186 / 25.370 / 27.034	cross-size transfer: ksum4 is incumbent at 2 sizes; ln N fitted on log2 r (slope 0.140) and extrapolated
`ksum4-n52`	ksum4	52	`prime-projected-abscissa:size=52,scan_cap=832`	`ksum:m=4`	16	16	97.863 / 97.959 / 98.821	cross-size transfer neighbourhood: N = 1.4 x 37 (ksum4 from k = 28)
`ksum4-n26`	ksum4	26	`prime-projected-abscissa:size=26,scan_cap=416`	`ksum:m=4`	16	16	8.303 / 8.819 / 13.460	cross-size transfer neighbourhood: N = 0.7 x 37 (ksum4 from k = 28)

arm	family	N	factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
ksum4-n19	ksum4	19	prime-projected-abscissa:size=19,scan_cap=304	ksum:m=4	16	16	9.045 / 10.523 / 23.829	cross-size transfer neighbourhood: N = 0.5 x 37 (ksum4 from k = 28)
ksum4-n74	ksum4	74	prime-projected-abscissa:size=74,scan_cap=1184	ksum:m=4	16	16	408.992 / 409.092 / 409.996	cross-size transfer neighbourhood: N = 2 x 37 (ksum4 from k = 28)
ksum3-n176	ksum3	176	prime-projected-abscissa:size=176,scan_cap=2816	ksum:m=3	16	16	156.453 / 156.760 / 159.522	cross-size transfer: ksum3-n11 is the k = 20 incumbent at N / N_a = 0.344
ksum3-n246	ksum3	246	prime-projected-abscissa:size=246,scan_cap=3936	ksum:m=3	16	16	426.529 / 427.077 / 432.011	cross-size transfer neighbourhood: N = 1.4 x 176 (ksum3 from k = 20)
mitmbatch-n512	mitmbatch	512	prime-projected-abscissa:size=512,scan_cap=8192	mitm-prime-batch:m=2	16	16	23.193 / 25.381 / 45.069	exploration: family mitmbatch untried at k = 32, N = 1 x N_a
mitmblocked-n512	mitmblocked	512	prime-projected-abscissa:size=512,scan_cap=8192	mitm-prime-blocked:m=2	16	16	23.193 / 25.381 / 45.069	exploration: family mitmblocked untried at k = 32, N = 1 x N_a



train screen

Selection

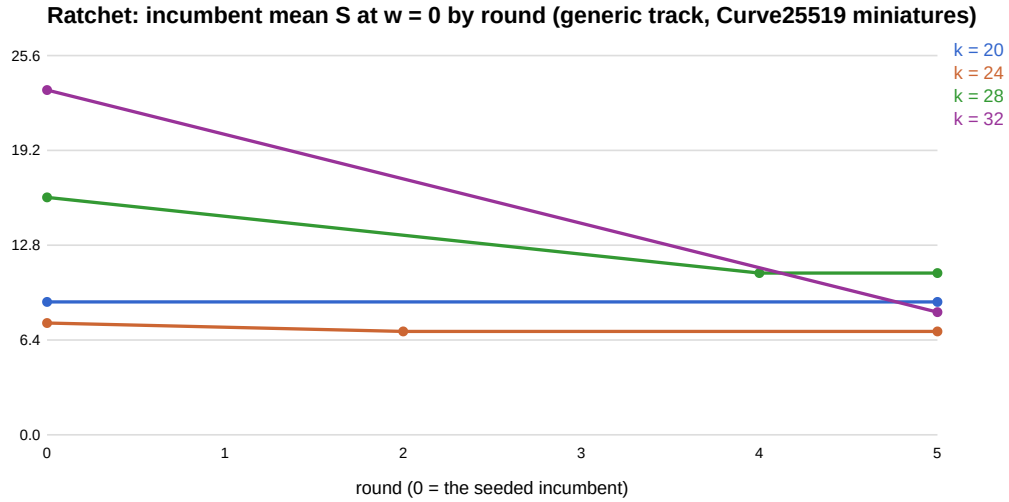
- k = 32: challenger `ksum4-n26` (Pareto alternative at w = 1: `ksum4-n26`)

Holdout sessions

k	session	run exit	verify exit	receipt sha256	levels
32	`ECBS1hbc72fda3e97f`	0	0	`66fa3e9a49eabcfa9b90bc60f28c909b548b10b516c674d8ca73594357a232b6`	{“verified”:480}

Ledger after this round

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
20	`ksum3-n11` (`prime-projected-abscissa:size=11,scan_cap=176` + `ksum:m=3`)	0	8.987 / 9.710 / 16.217	5.7x / 5.7x / 6.0x
24	`ksum4-n12` (`prime-projected-abscissa:size=12,scan_cap=192` + `ksum:m=4`)	2	6.985 / 7.514 / 12.271	6.0x / 5.9x / 5.6x
28	`ksum4-n21` (`prime-projected-abscissa:size=21,scan_cap=336` + `ksum:m=4`)	4	10.939 / 11.226 / 13.809	10.8x / 10.1x / 7.0x
32	`ksum4-n26` (`prime-projected-abscissa:size=26,scan_cap=416` + `ksum:m=4`)	5	8.296 / 8.811 / 13.446	9.1x / 8.8x / 7.4x



ratchet

Evidence and deviations

- ecbench binary sha256: `3fd22853485e388782ac0821295c56f81725fc832fd3610978be8088b5098bca`; `--cpus auto`, allow_busy = true; `verify --replay 8` per session; audit receipts in `analysis/`; sessions in `sessions/` (records in the artifact store when large).
- Route: the local route is an explicitly requested development screen, not the Iso Lab EKS default; counts are deterministic and replayed, wall time is not claimed. No row of this round enters the IC leaderboard or `benchmarks/README.md` (validation gate).
- Train selection is in sample (disclosed); the holdout is fresh (new `target_seed`, excluded exposures). Failed, timed-out and regressing arms are retained above.